



Full Length Article

Epileptic seizure detection using information Gain-Based hybrid Features: Deep Neural network and comparative Machine learning approaches

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ABSTRACT

Automatic detection of epileptic seizures is crucial in clinical diagnosis to enable early intervention and ensure patient safety. However, systematic comparisons across multi-class combinations and quantitative evaluation of discriminative features remain limited in the literature. This study aims to identify the most effective features for seizure detection and to develop a high-accuracy classification model. Statistical, spectral, and wavelet-based features from time, frequency, and time–frequency domains were selected using the Information Gain method, and four models were integrated into a hybrid framework. The approach was evaluated on 26 class combinations using Random Forest, Support Vector Machines, k-Nearest Neighbors, Gradient Boosting, and a Deep Neural Network. The proposed method achieved an average accuracy of 99%, with the Deep Neural Network reaching 99.69% in combinations including class E, demonstrating strong generalizability in multi-class scenarios. The main novelty of this work lies in combining Information Gain-based hybrid feature selection with a systematic multi-class analysis, a gap not fully addressed in previous studies. This approach enhances accuracy, interpretability, and generalizability, thereby contributing to improved clinical decision-making in epilepsy diagnosis.

1. Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures that result from abnormal, synchronized neuronal discharges in the brain. According to the World Health Organization (WHO) and recent international clinical guidelines, the diagnosis of epilepsy is established based on these recurrent, unprovoked seizures and the detection of abnormal electrical brain activity [1,2]. Electroencephalography (EEG) remains the clinical gold standard for this purpose, as it records brain activity with high temporal resolution and reveals abnormal wave patterns during both seizure and interictal periods [3]. Therefore, the present study is grounded in these globally accepted medical standards, using EEG as the primary diagnostic reference for the detection and analysis of epileptic seizures.

Timely detection of epileptic seizures is vital for improving patients' quality of life. For patients, it helps prevent injuries, supports treatment monitoring, and allows lifestyle adjustments. For clinicians, reliable detection provides objective diagnostic data and guides personalized treatment, drug dosage, and surgical decisions. For caregivers, it enables rapid intervention and ensures both physical and psychological safety. Overall, rapid and accurate seizure detection enhances patient

independence and reduces family anxiety, highlighting the need for reliable automatic detection systems.

Automatic detection of epileptic seizures requires a multidisciplinary approach due to the complex and nonstationary nature of EEG signals [4]. Time-domain analyses extract basic statistical features such as mean, variance, and energy to capture seizure-specific patterns, while frequency-domain methods such as Fourier or Welch transform identify abnormal low- and high-frequency components [5]. Time–frequency analyses, including short-time Fourier transform (STFT) and discrete wavelet transform (DWT), provide simultaneous insight into temporal and spectral changes, making them effective for transient seizure events [6,7]. In addition, similarity measures such as cross-correlation and dynamic time warping (DTW) characterize propagation patterns across electrodes, with DTW compensating for temporal misalignments and cross-correlation quantifying linear relationships [8,9].

Recently, hybrid approaches have become a major trend in seizure detection [10]. These methods integrate features from time, frequency, and time–frequency domains to improve classification accuracy [11–13]. Wavelet-based or other time–frequency features are commonly combined with machine learning algorithms such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forest (RF),

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k-Nearest Neighbors (k-NN), and Adaptive Neuro-Fuzzy Inference Systems (ANFIS) [14–17]. Deep learning methods, particularly fully connected Deep Neural Networks (DNN), have further advanced seizure detection, achieving high performance on large datasets [18]. By integrating diverse feature types and classifiers, hybrid frameworks provide more reliable and accurate outcomes, thereby enhancing the effectiveness of early seizure detection in clinical practice [19–21].

Studies in the literature on automatic epileptic seizure detection have reported high accuracies using hybrid approaches based on time, frequency, and time–frequency domain features [21]. However, most of these studies have examined only a limited number of class combinations or specific patient groups, which restricts their generalization capability. Moreover, comprehensive evaluations remain scarce regarding which features are truly discriminative in multi-class classification scenarios and how different class combinations affect model performance. Consequently, there is still a need for systematic investigations that compare performance across multiple class combinations and quantitatively assess the importance of features.

To address this gap, the present study applies an Information Gain (IG)-based feature selection method. Statistical features such as mean, variance, energy, skewness, and kurtosis are extracted from the time domain; power spectrum, spectral density, band energies, and entropy are derived from the frequency domain; and sub-band energies from wavelet transform are obtained from the time–frequency domain. These features are integrated into a hybrid framework. The framework aims to preserve the strengths of different models while selecting the most informative features through the IG method. A systematic classification analysis is then performed on 26 class combinations using both traditional machine learning algorithms (RF, SVM, k-NN, Gradient Boosting (GB)) and a fully connected DNN. This strategy enables the quantitative identification of discriminative features, clarifies how classifier performance varies across class settings, and provides a structured way to optimize seizure detection in multi-class scenarios.

The contribution of this study lies in expanding the scope of comparative analyses by systematically exploring multi-class combinations and evaluating feature importance in a quantitative manner. By integrating feature selection with both classical and deep learning classifiers, the framework offers insights into which features and algorithms are more effective under different conditions. In doing so, it aims to improve not only accuracy but also interpretability and generalizability, thereby establishing a useful reference for future research and supporting the development of clinically meaningful seizure detection systems.

Building upon these motivations, the following section introduces the proposed framework and details the dataset, feature extraction strategies, feature selection, and classifiers employed in this study.

2. Materials and methods

Let $X = \{x_1, x_2, \dots, x_N\}$ denote the set of EEG signal segments, where each $x_i \in \mathbb{R}^T$ represents a time series of length $T = 2048$. Each segment is associated with a class label $y_i \in \{A, B, C, D, E\}$ corresponding to normal or epileptic states.

The objective is to learn a mapping function

$$f: \mathbb{R}^T \rightarrow \{A, B, C, D, E\} \quad (1)$$

that maximizes the classification accuracy of unseen EEG segments. To achieve this, we define a feature extraction operator

$$\phi: \mathbb{R}^T \rightarrow \mathbb{R}^d \quad (2)$$

which maps the raw EEG segment into a d -dimensional feature space combining time, frequency, and time–frequency representations. Then, an IG-based feature selection process

$$\phi^*: \mathbb{R}^d \rightarrow \mathbb{R}^{d^*}, \quad d^* < d \quad (3)$$

is applied to retain only the most discriminative features. Finally, a classifier C is trained such that

$$\hat{y} = C(\phi^*(x)) \quad (4)$$

with the goal of minimizing the empirical risk

$$\min_C \frac{1}{N} \sum_{i=1}^N k(y_i, \hat{y}_i) \quad (5)$$

where $k(\cdot, \cdot)$ denotes the classification loss function.

The proposed framework for epileptic seizure detection, illustrated in Fig. 1, is designed to address the above classification problem. The system operates in four main stages:

- **Dataset Preparation:** Raw EEG signals from the Bonn dataset are segmented into equal-length windows to increase the number of training samples while preserving temporal information.
- **Feature Extraction (Models 1–4):** Complementary features are derived from multiple domains: Cross-correlation features derived from power spectral density (Model 1), frequency-domain similarity measures with reference PSDs (Model 2), wavelet-based statistical descriptors from Discrete Wavelet Transform sub-bands (Model 3), and time-domain statistical features such as Hjorth parameters and RMS values (Model 4).
- **Feature Selection via IG:** From the initially extracted 52 features, the IG criterion is used to select the most discriminative subset, ensuring that redundant or noisy attributes are discarded while maximizing class separability.
- **Classification:** The reduced feature set is provided to multiple classifiers (RF, SVM, k-NN, GB, DNN). The performance of these models is compared under 10-fold cross-validation to evaluate generalizability across binary and multi-class combinations.

By integrating diverse features and optimizing them through IG, the proposed system ensures both robustness and interpretability. This structured design effectively transforms the abstract mathematical formulation of seizure detection into a practical, clinically applicable framework.

2.1. EEG dataset

In this study, the electroencephalography (EEG) dataset provided by the Department of Epileptology at the University of Bonn, which is frequently referenced in the literature for epileptic seizure detection, was used. The dataset consists of five distinct subsets labeled A, B, C, D, and E, each recorded in a controlled environment, pre-labeled, and free of artifacts. Each subset contains 100 single-channel EEG recordings obtained from five different healthy subjects or patients with epilepsy. The recordings cover a duration of approximately 23.6 s with a sampling frequency of 173.61 Hz, offering high temporal resolution. In this study, the raw signals of the dataset were used without any preprocessing or filtering. The subsets represent recordings from healthy individuals with eyes open, healthy individuals with eyes closed, epileptic patients during seizure-free intervals, and epileptic patients during seizures from the epileptogenic zone. Owing to this structure, the dataset provides a balanced representation of both normal and pathological EEG characteristics and establishes a reliable basis for automatic epileptic seizure detection [22].

The original EEG recordings consist of 4096 samples, corresponding to approximately 23.6 s at a sampling frequency of 173.61 Hz. No filtering, normalization, or windowing operations were applied during preprocessing. However, to increase the number of samples in the dataset and enhance the learning capacity of the model, each recording was divided into two equal segments. Each segment thus contained 2048 samples, corresponding to approximately 11.8 s, thereby doubling the

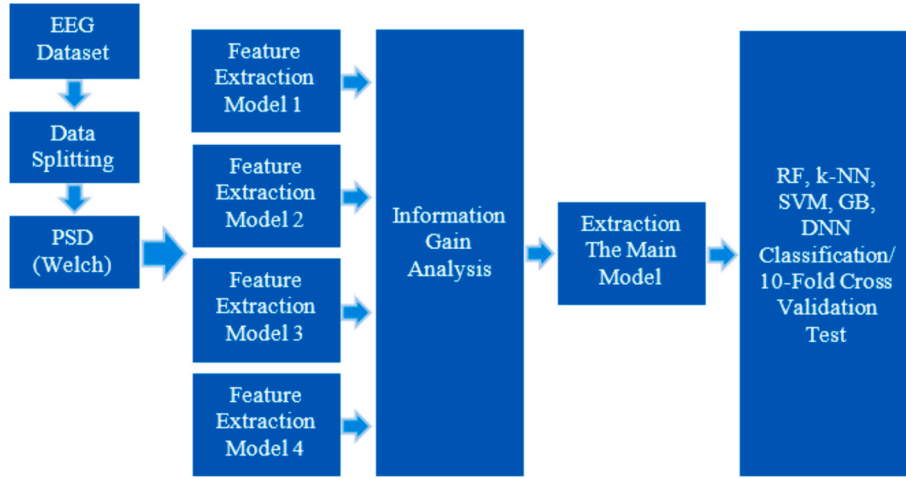


Fig. 1. Block diagram representation of the proposed system.

total number of available signals. This segmentation strategy preserved the continuity of the signals while enriching the information obtained from different temporal intervals. To avoid potential temporal redundancy or correlation between adjacent halves, segmentation was applied before the train/test division, and data partitioning was randomized at the segment level to minimize the likelihood of correlated samples appearing in both subsets. Consequently, the adopted segmentation approach improved data diversity and model generalizability without introducing bias or overestimation in performance.

Based on this dataset structure, the next step was to design feature extraction models capable of capturing the discriminative patterns of seizure and non-seizure EEG signals.

2.2. Feature extraction techniques

To obtain meaningful information from EEG signals and to reliably detect epileptic seizures, different feature extraction methods were applied. These methods aim to capture the characteristics of the signal in both the frequency and time domains, thereby revealing the complexity of the signal, the distribution of its energy, its waveform structure, and its morphological properties from different perspectives. In this study, four main methods were employed. These include cross-correlation features based on power spectral density, spectral features, wavelet-based features obtained through DWT and statistical parameters directly extracted from the time domain. The fundamental principles and selection criteria of each method are explained in detail below.

2.2.1. Cross-correlation based feature extraction – Model 1

In this method, spectral features were directly derived from the Power Spectral Density (PSD) of the EEG signals, which was estimated using the Welch method [23,24]. Unlike the cross-correlation-based approach that evaluates similarity with a reference spectrum, this model focuses on capturing intrinsic spectral characteristics of the signal itself. The PSD provides information about how the signal's energy is distributed across different frequency bands, thereby revealing frequency-domain patterns associated with normal and epileptic brain activity.

From the PSD representation, several statistical and spectral measures were extracted to characterize the distribution of spectral energy. These included the mean (μ), variance ($\text{Var}(x)$), standard deviation (σ), skewness (Skew), kurtosis (Kurt), signal energy (E), spectral entropy (H), and PSD-based cross-correlation measures, namely zero-lag correlation ($r(0)$) and maximum-lag correlation ($\rho_{\max}$). In these expressions, $x[n]$ denotes the EEG signal, $P_x[k]$ and $P_r[k]$ represent the PSDs of the test and reference signals, $\bar{P}_x$ and $\bar{P}_r$ are their mean PSD values, and $\bar{P}_x[k] =$

$P_x[k] / \sum_k P_x[k]$ is the normalized PSD used for entropy calculation. The complete mathematical definitions of all Model 1 features are summarized in Table 1.

According to IG analysis, the most discriminative features were identified as spectral entropy, spectral centroid, spectral flatness, and peak frequency. These features effectively highlight differences in the spectral signatures of epileptic versus non-epileptic EEG segments, allowing robust characterization of abnormal rhythmic activity. By emphasizing the frequency distribution dynamics, the spectral feature model provides a complementary perspective to cross-correlation analysis and contributes to improving the reliability of seizure detection.

2.2.2. Spectral feature extraction – Model 2

In this method, the frequency components of EEG signals were obtained using the Welch PSD estimation. The Welch method divides the signal into short segments, computes the periodogram of each segment, and averages them, thereby reducing the variance of spectral estimation and minimizing the effect of noise. The PSD of each EEG signal was compared with the mean reference PSD of its corresponding class, and from this comparison seven distinct features reflecting frequency-domain differences were extracted. These features include the Euclidean distance between the signal PSD and the reference PSD (general spectral dissimilarity), spectral entropy difference (regularity of frequency distribution), maximum power difference between the two PSDs, maximum power density indicating the dominant frequency, frequency variance of the power spectrum, Kullback–Leibler divergence (statistical similarity of two distributions), and median frequency

Table 1
Mathematical definitions of the features for Model 1.

Feature	Mathematical Definition
Mean	$\mu = \frac{1}{N} \sum_n x[n]$
Variance	$\text{Var}(x) = \frac{1}{N} \sum_n (x[n] - \mu)^2$
Standard Deviation	$\sigma = \sqrt{\text{Var}(x)}$
Skewness	$\text{Skew} = \frac{\frac{1}{N} \sum (x[n] - \mu)^3}{\sigma^3}$
Kurtosis	$\text{Kurt} = \frac{\frac{1}{N} \sum (x[n] - \mu)^4}{\sigma^4}$
Energy	$E = \sum_n x[n]^2$
Spectral Entropy	$H = -\sum_k \bar{P}_x[k] \log(\bar{P}_x[k])$
Zero Lag (PSD)	$r(0) = \sum_k (P_x[k] - \bar{P}_x)(P_r[k] - \bar{P}_r)$
Maximum Lag (PSD)	$\rho_{\max} = \underset{l}{\text{argmax}} \quad r(l) $

representing the balance of energy distribution [25].

Table 2 presents the mathematical definitions of the features extracted for Model 2. In this table, $P_x[k]$ denotes the PSD of the current EEG signal at frequency bin k , $P_r[k]$ represents the mean reference PSD of the corresponding class, f_k is the frequency at bin k , $\bar{f}$ is the mean frequency, and K is the total number of frequency bins.

IG analysis revealed that maximum spectral deviation and Kullback–Leibler divergence were the most discriminative features. Maximum spectral deviation directly captures pronounced power differences in lower frequency bands during epileptic seizures when compared to the reference PSD, whereas Kullback–Leibler divergence quantifies the statistical distance between two spectral distributions, thereby emphasizing changes in energy allocation. Together, these two parameters effectively measure differences in spectral shape and energy distribution, providing high discriminative IG for epileptic seizure detection.

2.2.3. Discrete wavelet transform based feature extraction- model 3

In this method, EEG signals were analyzed using the discrete wavelet transform (DWT), which provides both time and frequency resolution. Specifically, the Daubechies 4 (Db4) wavelet was employed, as it is effective in capturing abrupt transitions and oscillations in EEG signals. Let $c_s[m]$ denote the wavelet coefficients of sub-band $s \in \{A4, D4, D3, D2\}$. For each sub-band, two basic amplitude measures were directly computed. The minimum coefficient value $\min_m c_s[m]$ and the maximum coefficient value $\max_m c_s[m]$. In addition to these, four statistical descriptors, mean, standard deviation, skewness, and kurtosis, were also extracted from each sub-band. Their mathematical formulations are identical to those used in Model 1 and referred to Table 1 for the exact definitions of these expressions. Collectively, these six measures were calculated for each wavelet sub-band $s \in \{A4, D4, D3, D2\}$, yielding a total of 24 features for Model 3. This multiscale approach effectively reflects the energy and structural differences observed in the frequency bands of epileptic seizures, thereby providing a strong foundation for analysis [26,27].

Based on IG analysis, the standard deviation values in the higher-frequency sub-bands (D4, D3, and D2) and the kurtosis parameters in the D3 and D4 sub-bands were found to have high discriminative power. The standard deviation in the D4 and D3 sub-bands indicates increased frequency fluctuations in seizure EEGs, while the kurtosis parameters reflect the intensity of sharp peaks and outliers in the energy distribution. Furthermore, the standard deviation of the A4 sub-band, which represents the lowest frequency components, captures the variability of long-term fluctuations and thus differentiates seizure from non-seizure states. These selected wavelet features represent the statistical structure of EEG signals across frequency bands in detail, providing high IG for epileptic seizure detection.

2.2.4. Time-domain feature extraction – Model 4

In this method, EEG signals were analyzed directly in the time domain. Time-domain features provide direct information about the

amplitude distribution, fluctuation intensity, and morphological complexity of the signal. Sudden amplitude increases, sharp peaks, and irregular oscillations observed during epileptic seizures can be effectively represented through such features. For each signal, a total of 12 features were calculated, including mean, median, mode, standard deviation, skewness, kurtosis, root mean square (RMS) value, the sum of absolute differences between successive samples, zero-crossing rate, and Hjorth parameters (activity, mobility, and complexity) [28,29].

Table 3 presents the mathematical definitions of the features extracted for Model 4. In this table $x(t)$ denotes the time-domain EEG signal of interest and N is the total number of samples. The mean represents the average amplitude of the signal and is calculated using the same formula as in Table 1. Similarly, the variance, standard deviation, skewness, and kurtosis are computed using the formulas in Table 1, providing measures of dispersion, asymmetry, and peakedness respectively. The median corresponds to the value separating the lower and upper halves of the amplitude distribution, while the mode represents the most frequently occurring amplitude value. The Hjorth activity parameter is defined as the variance of $x(t)$ and captures the overall signal power, reflecting the intensity of signal fluctuations. Together with mobility and complexity, it characterizes the morphological dynamics of the EEG waveform.

According to the IG analysis, several time-domain features were identified as highly discriminative: the Hjorth complexity parameter, which measures waveform complexity; the sum of absolute differences between successive samples, which captures the “roughness” of the signal; Hjorth mobility, which reflects sensitivity to the mean frequency content; standard deviation, which represents amplitude variability; the RMS value, which indicates signal power; the zero-crossing rate, which measures the frequency of sign changes; and Hjorth activity, a variance-based indicator of total signal energy. These features comprehensively and complementarily represent the differences in fluctuation, energy, and morphological complexity between seizure and non-seizure EEG signals, thereby making a significant contribution to classification performance.

2.3. Information gain analysis (IGA)

IGA is a widely used metric in machine learning and data mining to measure the ability of a feature to reduce uncertainty in class labels. Essentially, IGA determines the extent to which a feature decreases the entropy value of the class distribution. A high IG indicates that the feature can better discriminate between classes, thereby contributing positively to classification performance. Therefore, IGA serves as an important tool in selecting the most distinctive and meaningful features from a large set of candidates. For high-dimensional datasets, IG values are referenced during the feature selection stage to reduce model complexity and improve generalization capability [30].

2.3.1. Ig-based feature selection and main model formation

The Bonn University EEG dataset, widely used for automatic epileptic seizure detection, consists of five classes (A–E), each representing different neurological conditions. In this study, various class

Table 2
Mathematical definitions of the features for Model 2.

Feature	Mathematical Definition
Peak Frequency	$f_{\text{peak}} = \text{argmax}_f P_x(f)$
Median Frequency	$\sum_{f < f_{\text{med}}} P_x(f) = \frac{1}{2} \sum_f P_x(f)$
KL Divergence	$D_{\text{KL}} = \sum_k P[k] \log \frac{P[k]}{Q[k]}$
Frequency Variance	$\sigma_f^2 = \sum_k (f_k - \bar{f})^2 P_x[k]$
Entropy Difference	$\Delta H = -\sum_k P_x[k] \log P_x[k] + \sum_k P_r[k] \log P_r[k]$
Euclidean Distance	$d = \sqrt{\sum_k (P_x[k] - P_r[k])^2}$
Mean Spectral Deviation	$\Delta_{\text{mean}} = \frac{1}{K} \sum_{k=1}^K (P_x[k] - P_r[k])$

Table 3
Mathematical definitions of the features for Model 4.

Feature	Mathematical Definition
RMS	$\text{RMS} = \sqrt{\frac{1}{N} \sum x[n]^2}$
Zero-Crossing Rate	$\text{ZCR} = \frac{1}{N-1} \sum b1(x[n]x[n+1] < 0)$
Hjorth Activity	$\text{Activity} = \text{Var}(x)$
Hjorth Mobility	$\text{Mobility} = \sqrt{\frac{\text{Var}(x')}{\text{Var}(x)}}$
Hjorth Complexity	$\text{Complexity} = \frac{\text{Mobility}(x')}{\text{Mobility}(x)}$

combinations were created to evaluate classification performance. If only a single class combination had been considered, feature selection could have been based on the top IG values within each model. However, since 26 class combinations were analyzed, the main model was instead constructed from features achieving the highest average IG across all combinations, ensuring generalizability. However, since 26 different class combinations were analyzed, feature selection based on a single combination would not be suitable for all cases.

Therefore, the data files containing the feature vectors generated for each model were processed separately. The workflow of the program is as follows:

1. **Data and Feature Reading:** Each data file contained EEG features and class labels corresponding to a specific class combination. These files were read individually, and the feature names and data matrix were extracted.
2. **IG Calculation:** The IG value was calculated for each feature. This calculation quantitatively measures the contribution of a feature to class discrimination and is obtained using the formula $H(Y) - H(Y|X)$, where $H(Y)$ denotes the entropy of the class labels and $H(Y|X)$ represents the conditional entropy given the feature.
3. **Aggregation Across All Combinations:** The IG values obtained for each class combination were collected into a matrix. Using this matrix, the average IG across all combinations was computed.
4. **Selection of Main Model Features:** Based on the average IG values across all combinations, the most informative features were identified. This approach, instead of relying on a single combination, enables the construction of a main model that is more generalizable and stable by incorporating a multi-combination perspective.

Through this method, rather than selecting the optimal model for each class combination separately, a global IG-based feature set covering all combinations was derived. This approach is expected to enhance the generalizability of feature selection and improve the stability of the model, particularly in multi-class EEG datasets where class combinations differ.

The choice of IG as the primary feature selection criterion was both theoretically and empirically motivated. Theoretically, IG quantifies the expected reduction in class entropy, providing a direct and model-independent measure of how much uncertainty is reduced when a given feature is known. This makes IG particularly suitable for EEG data, which are inherently nonlinear and nonstationary, because IG does not assume any linear relationship between features and class labels. Empirically, IG offers a stable and computationally efficient ranking of features across multiple class combinations, allowing systematic comparison of large feature spaces without requiring parameter tuning or iterative optimization. These properties make IG an appropriate choice for identifying the most discriminative features under the multi-class structure employed in this study.

Although IG itself does not explicitly remove redundancy among features, this limitation was mitigated through a two-step filtering strategy. First, features with closely similar definitions or strong pairwise correlations (e.g., overlapping spectral or statistical measures) were manually reviewed and consolidated before IG computation. Second, when aggregating IG values across 26 class combinations, features showing unstable or marginal contribution were discarded. This hybrid procedure ensured that the final feature subset remained both informative and minimally redundant, achieving a balance between interpretability, discriminative power, and computational efficiency.

The reduced feature set obtained through IG formed the basis of the main model, which was subsequently evaluated using multiple classification algorithms.

2.4. Classifiers and performance metrics

Comparing the performance of different classification algorithms in

epileptic seizure detection is of critical importance for model selection and practical applicability. In this study, both classical machine learning methods and a deep learning model were employed to evaluate the effectiveness of discriminative EEG features.

The RF algorithm is an ensemble method composed of multiple decision trees and is preferred due to its high accuracy and ability to reduce the risk of overfitting. RF performs classification based on majority voting among trees trained on different subsets of the data, thereby producing stable results [31]. In this study, it was used to capture the complex structure of EEG data and to enhance generalization performance.

k-NN algorithm is a simple yet effective classification method that assigns data points according to the class of their k nearest neighbors [14]. Here, k was set to 3, and k-NN was used as a reference model since it demonstrates intuitive performance, particularly on small datasets.

SVM are powerful linear and non-linear classifiers that focus on finding the maximum margin between classes [14]. They were selected in this study for their ability to perform effective discrimination in high-dimensional and complex data such as EEG, with kernel functions providing flexible modeling capability.

GB is an ensemble method in which weak classifiers are trained sequentially, with errors reduced in a stepwise manner [15]. Due to its ability to enhance learning capacity and improve classification accuracy, it was used in this study to strengthen discrimination in complex EEG signals.

Finally, DNNs are powerful deep learning-based models that can capture complex nonlinear relationships in EEG data through multiple fully connected layers and regularization mechanisms [17]. In this study, DNN was compared with traditional machine learning methods to investigate the contribution of deep learning to classification performance.

The performance of the models was evaluated using metrics such as accuracy, precision, recall, and F1-score, which allow for a more comprehensive analysis by assessing performance from different perspectives. Accuracy indicates the overall proportion of correctly classified samples, while precision measures false positives and recall addresses false negatives. The F1-score provides a balanced single-value summary of precision and recall, thereby reflecting overall model performance more accurately [32]. These metrics were used to comprehensively assess the reliability and effectiveness of classifiers in critical applications such as epileptic seizure detection.

In this study, the hyperparameters of all classifiers were determined not solely based on literature recommendations or default values but through a structured tuning procedure. For this purpose, a two-stage strategy was applied. First, a coarse grid search was conducted over the hyperparameter ranges commonly used in the literature, and then a more refined search was performed within narrower intervals around the values that showed promising preliminary performance. Throughout the tuning process, ten-fold cross-validation on the training data was employed, and the average accuracy across folds was used as the selection criterion.

For the RF classifier, the number of trees between 50 and 300, the maximum depth including unrestricted values and depths from 10 to 50, and the minimum number of samples per leaf between 1 and 5 were evaluated. The configuration consisting of 100 trees, unrestricted depth, and one sample per leaf produced the most stable results. For the SVM classifier, the C parameter between 0.1 and 10 and RBF kernel gamma values including automatic estimation and values between 0.0001 and 0.1 were tested, and C equal to 1 with automatic gamma achieved the best balance between bias and variance. For k-NN, values of k between 1 and 10 and distance metrics including Euclidean, cosine, and Minkowski were examined. The configuration using k equal to 3 with Euclidean distance provided the most stable performance. For the GB model, learning rates between 0.01 and 0.2, tree counts between 50 and 200, and maximum depths between 2 and 5 were analyzed. The configuration of 100 trees, a learning rate of 0.1, and a depth of 3 yielded the most

balanced results.

In addition to classical machine learning classifiers, a fully connected DNN was implemented to assess the contribution of deep learning to the classification performance. The architecture was designed based on systematic preliminary experiments that balanced model capacity and generalization while avoiding overfitting, considering the dimensionality of the feature vectors. The network consisted of a feature input layer matching the number of extracted attributes, followed by three dense layers with 256, 128, and 64 neurons, respectively. Each fully connected layer was followed by batch normalization to stabilize internal covariate shift, a ReLU activation function, and dropout regularization with rates of 0.40, 0.30, and 0.20 to prevent overfitting by reducing co-adaptation of neurons.

The final layer was a fully connected softmax output layer whose number of neurons equaled the number of target classes. Model training was conducted using the Adam optimizer with an initial learning rate of 0.001, a mini-batch size of 32, and a maximum of 100 epochs. To ensure reproducibility and proper monitoring of generalization, the available data were partitioned in two stages: first, 90% of the samples were used for training and 10% for testing, and second, an additional 20% of the training subset was randomly allocated for validation. All feature vectors were normalized using z-score statistics computed from the training portion only. The validation loss was monitored during training to mitigate overfitting.

This architectural configuration was chosen because deeper or wider networks yielded noticeable overfitting in preliminary trials, while shallower networks were insufficient to capture the nonlinear relationships between the extracted features. The selected design therefore represents a principled balance between expressive power and robustness, enabling DNN to achieve stable performance across different class combinations.

Having established the dataset, features, and classification framework, the next section reports the experimental results obtained from their application.

3. Experimental findings

In this study, publicly available EEG datasets from Bonn University

were used for epileptic seizure detection. All analyses and classifications were implemented using the MATLAB programming language. The experimental procedures were carried out on a computer equipped with an Intel i5 12th generation processor and 16 GB of RAM. Data loading, feature extraction, IG calculations, and model training processes were tested in this environment. This hardware and software infrastructure enabled the efficient processing of multiple models and class combinations employed in the study.

A total of 52 features were extracted using four different models. Considering the five classes included in the Bonn University EEG dataset, 26 distinct class combinations were generated. For each of these combinations, IG values were calculated for all features, and the features with the highest average IG values were identified. In Table 4, the features for each model are presented by considering only their own IG values, thereby reflecting the results specific to each model within the five classes of the Bonn dataset. This distinction highlights that Table 4 shows model-based IG results for reference only, while Table 5 represents the final feature selection of the main model, derived from the average IG values across all class combinations. Consequently, the feature vector of the main model was constructed through an optimized selection process that considered the performance of features across all class combinations and models.

Table 4 presents the average IG values of features within each individual model, serving only to illustrate model-specific tendencies. These values were not directly used in the construction of the main model but rather provided a comparative perspective. The results demonstrate that each model emphasizes different types of information extracted from the dataset. In Model 1, statistical distribution features such as Entropy (0.697) and Skewness (0.614) were most prominent. In Model 2, frequency-domain features such as Peak Frequency (0.512) and Kullback–Leibler Divergence (0.221) achieved higher importance. Model 3 primarily ranked wavelet transform-based statistics such as D4 Maximum (0.440) and D4 Standard Deviation (0.411) at the top. Model 4 highlighted time-domain complexity measures, with Hjorth Complexity (0.796) and Hjorth Mobility (0.658), as well as energy-based features producing strong results. These findings indicate that different model configurations tend to capture distinct types of information from the data, suggesting that feature selection may require model-specific

Table 4

Comparative representation of IG values of the four models. (Average IG values across all class combinations were used, and in each row the features with the highest IG values for the corresponding model and their respective scores are listed. The symbol “–” denotes that no corresponding feature was present in that model for the given row).

#	Model 1	M1 IG	Model 2	M2 IG	Model 3	M3 IG	Model 4	M4-IG
1	Entropy	0.697	Peak Freq.	0.512	A4 Std	0.445	Hjorth Comp.	0.796
2	Skewness	0.614	KL Divergence	0.221	D4 Max	0.440	Hjorth Mobility	0.658
3	Variance	0.496	Freq. Var.	0.200	D4 Std	0.411	RMS	0.495
4	Std. Dev.	0.473	Euclidean Dist.	0.142	D4 Min	0.394	Std. Dev.	0.492
5	Mean	0.465	Mean Spc. Dev.	0.131	A4 Min	0.374	Zero Cross. Rate	0.434
6	Energy	0.412	Entropy Diff.	0.096	D3 Std	0.363	Sum Difference	0.382
7	Kurtosis	0.358	Median Freq.	0.071	D3 Min	0.346	Skewness	0.232
8	Zero Lag	0.245	–	–	A4 Max	0.345	Hjorth Activity	0.231
9	Max Lag	0.202	–	–	D3 Max	0.318	Median	0.151
10	–	–	–	–	D2 Max	0.270	Mode	0.113
11	–	–	–	–	D2 Min	0.247	Kurtosis	0.082
12	–	–	–	–	D2 Std	0.232	Mean	0.070
13	–	–	–	–	D3 Mean	0.161	–	–
14	–	–	–	–	A4 Mean	0.105	–	–
15	–	–	–	–	D4 Mean	0.088	–	–
16	–	–	–	–	D2 Kurtosis	0.084	–	–
17	–	–	–	–	D2 Mean	0.078	–	–
18	–	–	–	–	D3 Kurtosis	0.074	–	–
19	–	–	–	–	D2 Skewness	0.070	–	–
20	–	–	–	–	D4 Kurtosis	0.061	–	–
21	–	–	–	–	D3 Skewness	0.054	–	–
22	–	–	–	–	D4 Skewness	0.052	–	–
23	–	–	–	–	A4 Skewness	0.027	–	–
24	–	–	–	–	A4 Kurtosis	0.011	–	–

Table 5

The 20 highest-ranked features selected from the first four models and used in the classifications of the main model (Average IG value across all combinations were used).

#	Description	Source Model	IG Value
1	Hjorth Complexity	Model4	0.795
2	Level 4 Standard Deviation	Model3	0.730
3	Total Difference	Model4	0.704
4	Hjorth Mobility	Model4	0.653
5	Level 3 Standard Deviation	Model3	0.616
6	Level 2 Standard Deviation	Model3	0.609
7	Standard Deviation	Model4	0.598
8	RMS Value	Model4	0.568
9	Cross-Correlation Maximum Lag	Model1	0.483
10	Level 4 Energy	Model3	0.432
11	Zero Crossing Rate	Model4	0.432
12	Level 1 Standard Deviation	Model3	0.431
13	Level 3 Entropy	Model3	0.423
14	Hjorth Activity	Model4	0.420
15	Level 4 Entropy	Model3	0.404
16	Level 3 Energy	Model3	0.385
17	Bhattacharyya Distance	Model2	0.377
18	Cross-Correlation Zero Lag	Model1	0.363
19	Spectral Energy Difference	Model2	0.361
20	Kullback-Leibler Divergence	Model2	0.337

strategies.

In Table 5, the 20 most important features are listed. These were selected by averaging IG values across all class combinations, independent of the rankings shown in Table 4. Thus, Table 5 reflects the optimized feature vector of the main model, not a direct selection from the four sub-models. Among these, Hjorth Complexity (Model 4, 0.795), Level 4 Standard Deviation (Model 3, 0.730), and Total Difference (Model 4, 0.704) achieved the highest IG values, standing out as the features contributing most strongly to the discriminative power of the dataset.

Notably, the Hjorth parameters from Model 4 and the wavelet-based statistics from Model 3 appear predominantly in the list, indicating that these two models provide strong complementary elements in feature selection. In addition, cross-correlation-based features from Model 1 and spectral features from Model 2 are also included, supporting the idea that integrating diverse information sources from different model types can enhance classification performance.

To provide a clearer understanding of the discriminative capacity of each feature extraction strategy, the average IG values of Models 1–4 are summarized in Fig. 2. This figure highlights the relative contributions of time-domain, frequency-domain, wavelet-based, and cross-correlation

features to the overall classification performance. By visualizing the average IG values, it becomes evident which features carry higher discriminative power across all class combinations, thereby guiding the construction of the main hybrid model. The figure thus serves as an important reference point in demonstrating how diverse feature extraction techniques complement each other and why the integration of multi-domain features is essential for achieving robust and generalizable epileptic seizure detection.

The results presented in Table 6 and Table 7 provide a detailed illustration of the accuracy performances of four different models (M1–M4) across multiple class combinations using five classifiers. Examination of the overall trend reveals that the SVM and DNN algorithms, particularly in binary class combinations, achieved accuracy rates close to 100%, thereby demonstrating superior performance. The RF classifier also yielded high accuracy in both binary and multiclass scenarios, producing nearly perfect results especially in combinations such as AE, AD, CE, and BE. While the k-NN algorithm with the parameter $k = 3$ produced satisfactory outcomes in binary classifications, a noticeable decline in performance was observed as the number of classes increased; this suggests that its discriminative capacity may be limited in complex and high-dimensional data structures. GB generally maintained high accuracy levels and, in certain combinations, produced results approaching those of SVM and RF.

When comparing the models, it is observed that Model 2 demonstrated balanced and consistently high performance with most classifiers, whereas Model 4 produced strong results when its time-domain features were combined with DNN and SVM. Model 1, based on cross-correlation features, showed particularly good compatibility with RF and DNN. Model 3, while achieving high accuracy in certain class combinations with its wavelet-based statistical features, remained behind Model 2 and Model 4 in terms of overall average accuracy.

Considering the average accuracies across all classifiers, DNN and SVM generally achieved results in the range of 95–98%, while RF produced approximately 94%. Although k-NN yielded lower average accuracy compared to other methods, GB maintained consistently high accuracy levels. These findings indicate that DNN, as a deep learning-based approach and SVM, with its strong margin-based separation capability, are the most reliable classifiers for epileptic seizure detection, while Model 2 and Model 4 provided more generalizable outcomes across different class combinations in terms of feature extraction strategies.

The results presented in Table 8 and Table 9 comprehensively illustrate the performance of the main model, constructed using IG-based feature selection, across different class combinations and five

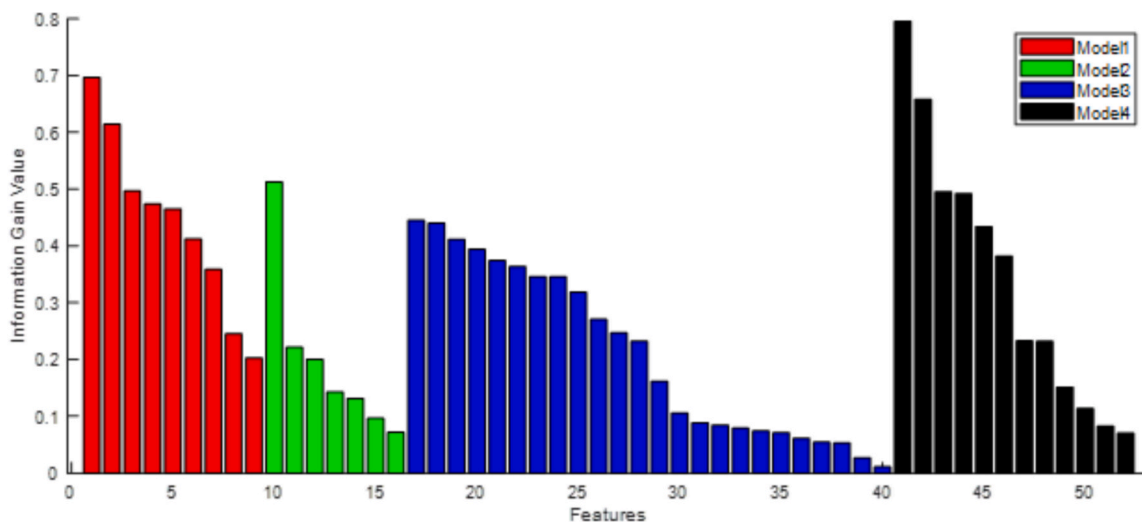
**Fig. 2.** Average IG values of Models 1, 2, 3, and 4.

Table 6

Accuracy values for multi-class classification obtained using four models with RF and SVM classifiers.

Class	RF				SVM			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
AB	0.937	0.975	0.975	0.887	1	1	0.887	0.875
ABC	0.883	0.95	0.883	0.875	1	0.958	0.85	0.941
ABCD	0.925	0.943	0.787	0.85	0.987	0.9	0.75	0.825
ABCDE	0.885	0.95	0.84	0.8	0.96	0.91	0.8	0.81
ABCE	0.968	0.956	0.906	0.918	0.993	0.975	0.925	0.906
ABD	0.958	0.983	0.908	0.841	1	0.983	0.9	0.966
ABDE	0.956	0.987	0.943	0.943	1	0.937	0.95	0.881
ABE	0.975	0.975	0.916	0.933	0.991	0.941	0.925	0.933
AC	1	0.9	0.95	0.987	1	0.95	0.975	0.987
ACD	0.883	0.891	0.858	0.808	0.975	0.916	0.825	0.741
ACDE	0.925	0.893	0.887	0.868	0.981	0.968	0.856	0.868
ACE	0.975	0.966	0.95	0.975	1	0.95	0.95	0.95
AD	1	0.987	0.95	0.925	1	1	0.925	0.9
ADE	0.991	0.975	0.933	0.941	0.991	0.991	0.983	0.941
AE	0.987	1	1	1	1	1	1	1
BC	0.987	1	0.912	0.975	1	0.937	0.95	0.975
BCD	0.95	0.941	0.808	0.908	0.958	0.858	0.808	0.833
BCDE	0.906	0.931	0.887	0.893	0.962	0.937	0.881	0.881
BCE	0.966	0.966	0.975	0.975	0.991	0.991	0.908	0.991
BD	0.987	1	0.962	0.975	1	0.975	1	0.987
BDE	0.975	0.975	0.958	0.966	0.991	0.975	0.983	0.966
BE	0.975	1	0.975	0.95	0.987	1	1	0.975
CD	0.775	0.937	0.737	0.725	0.862	0.9	0.687	0.812
CDE	0.858	0.941	0.816	0.808	0.933	0.941	0.791	0.833
CE	0.987	1	0.975	0.975	1	1	0.975	0.975
DE	0.975	0.987	0.987	0.962	0.987	1	1	0.937
Averaged	0.94	0.96	0.91	0.91	0.98	0.95	0.90	0.91

Table 7

Accuracy values for multi-class classification obtained using four models with k-NN (k = 3), GB and DNN classifiers.

Class	k-NN (k = 3)				GB				DNN			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
AB	0.987	0.937	0.8	0.837	0.987	0.987	0.9	0.862	1	1	0.837	0.887
ABC	1	0.9	0.825	0.925	0.991	0.933	0.95	0.916	0.991	0.975	0.916	0.95
ABCD	0.95	0.8	0.743	0.762	0.881	0.856	0.731	0.75	0.943	0.925	0.837	0.781
ABCDE	0.935	0.83	0.62	0.77	0.855	0.895	0.8	0.82	0.915	0.855	0.83	0.815
ABCE	0.981	0.893	0.656	0.868	0.987	0.937	0.906	0.937	0.993	0.918	0.856	0.925
ABD	0.983	0.941	0.841	0.883	0.983	0.941	0.933	0.925	1	0.775	0.875	0.941
ABDE	0.981	0.937	0.631	0.875	0.968	0.975	0.937	0.881	1	0.937	0.918	0.925
ABE	0.966	0.941	0.7	0.875	0.95	0.983	0.925	0.958	0.991	0.983	0.925	0.933
AC	1	0.912	0.95	0.975	1	0.925	0.987	0.987	1	0.975	0.975	0.962
ACD	0.891	0.875	0.733	0.808	0.95	0.875	0.808	0.841	0.958	0.883	0.866	0.841
ACDE	0.931	0.862	0.706	0.8	0.943	0.918	0.862	0.868	0.956	0.881	0.806	0.85
ACE	1	0.95	0.758	0.958	0.975	0.925	0.983	0.966	1	0.958	0.975	0.991
AD	1	0.937	0.925	0.95	1	0.962	1	0.937	1	0.975	1	0.987
ADE	1	0.95	0.808	0.941	1	0.991	0.975	0.958	1	0.966	0.966	0.983
AE	1	0.987	0.95	0.987	1	1	1	1	1	1	1	0.987
BC	1	0.925	0.962	0.95	0.987	1	0.975	1	1	0.975	0.95	0.975
BCD	0.916	0.858	0.808	0.808	0.85	0.95	0.825	0.858	0.925	0.825	0.841	0.875
BCDE	0.906	0.787	0.8	0.843	0.881	0.912	0.793	0.887	0.962	0.918	0.881	0.85
BCE	0.983	0.958	0.825	0.958	0.966	0.983	0.958	0.983	0.983	0.975	1	0.975
BD	1	0.975	0.95	0.975	1	0.987	0.962	1	1	0.937	0.962	0.987
BDE	1	0.925	0.8667	0.958	0.991	0.983	0.958	0.958	1	0.975	0.975	0.983
BE	0.987	1	0.975	0.95	0.95	1	1	0.987	0.975	1	0.987	1
CD	0.9	0.825	0.775	0.775	0.837	0.937	0.8	0.725	0.912	0.837	0.7	0.737
CDE	0.891	0.783	0.766	0.766	0.9	0.933	0.758	0.808	0.875	0.875	0.766	0.875
CE	0.987	1	0.962	0.987	1	1	0.975	0.987	1	1	1	1
DE	1	0.975	0.937	0.975	0.987	0.937	1	0.962	1	1	0.975	0.987
Averaged	0.96	0.91	0.81	0.89	0.95	0.95	0.91	0.91	0.97	0.93	0.90	0.92

separate classifiers. The results strongly support the model's generalizability and stable performance.

Among all classifiers, SVM and DNN stood out by achieving accuracy values up to 99% in nearly all class combinations. These two methods produced almost error-free results in binary class scenarios and maintained high accuracy even in multiclass situations. The RF algorithm also demonstrated strong performance, achieving 100% accuracy in combinations such as AE, ACE, AD, ADE, and BE, thereby serving as a

powerful alternative. GB provided results close to RF in most combinations, occasionally reaching 100% as well. k-NN exhibited more consistent performance when applied with the main model, though it lagged other classifiers in highly complex multiclass combinations.

Analysis of class-specific tendencies revealed that combinations involving epileptic seizure classes, characterized by distinctive frequency and time-domain features, consistently achieved high accuracy rates. For instance, in combinations such as AE, AD, and ADE, all

Table 8

Multi-class accuracy (Acc.), precision (Prec.), recall (Rec.), and F1-score (F1) values for RF and SVM classifier combinations within the main model.

Class	RF				SVM			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
AB	0.975	0.976	0.974	0.974	1	1	1	1
ABC	0.958	0.959	0.959	0.958	0.991	0.991	0.993	0.992
ABCD	0.95	0.951	0.952	0.951	0.968	0.971	0.970	0.970
ABCDE	0.93	0.930	0.936	0.930	0.965	0.96	0.968	0.966
ABCE	0.987	0.987	0.987	0.987	0.993	0.99	0.993	0.993
ABD	0.975	0.974	0.974	0.974	0.991	0.991	0.993	0.992
ABDE	0.981	0.982	0.9811	0.981	1	1	1	1
ABE	0.975	0.975	0.975	0.975	1	1	1	1
AC	0.987	0.989	0.985	0.987	1	1	1	1
ACD	0.933	0.934	0.934	0.934	0.958	0.963	0.960	0.960
ACDE	0.9563	0.951	0.9547	0.951	0.943	0.943	0.946	0.943
ACE	1	1	1	1	1	1	1	1
AD	0.987	0.989	0.984	0.986	1	1	1	1
ADE	1	1	1	1	1	1	1	1
AE	1	1	1	1	1	1	1	1
BC	1	1	1	1	0.987	0.9884	0.986	0.987
BCD	0.95	0.955	0.9538	0.954	0.983	0.983	0.984	0.983
BCDE	0.956	0.958	0.9602	0.958	0.981	0.984	0.980	0.981
BCE	0.975	0.973	0.978	0.975	1	1	1	1
BD	0.987	0.988	0.986	0.987	1	1	1	1
BDE	1	1	1	1	0.991	0.99	0.989	0.990
BE	1	1	1	1	1	1	1	1
CD	0.937	0.942	0.931	0.935	0.975	0.974	0.974	0.974
CDE	0.9083	0.918	0.9101	0.913	0.991	0.988	0.993	0.990
CE	1	1	1	1	1	1	1	1
DE	0.987	0.987	0.987	0.987	1	1	1	1
Averaged	0.973	0.974	0.973	0.973	0.989	0.989	0.989	0.989

Table 9

Multi-class accuracy (Acc.), precision (Prec.), recall (Rec.), and F1-score (F1) values for k-NN (k = 3), GB and DNN classifier combinations within the main model.

Class	k-NN (k = 3)				GB				DNN			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
AB	0.95	0.95	0.954	0.949	1	1	1	1	1	1	1	1
ABC	0.975	0.977	0.974	0.975	0.975	0.972	0.974	0.973	0.983	0.985	0.984	0.984
ABCD	0.9	0.902	0.9019	0.901	0.956	0.956	0.956	0.956	0.968	0.967	0.968	0.967
ABCDE	0.915	0.904	0.9051	0.903	0.95	0.950	0.95	0.949	0.965	0.964	0.962	0.962
ABCE	0.9563	0.955	0.955	0.955	0.968	0.970	0.969	0.969	1	1	1	1
ABD	0.9833	0.981	0.986	0.983	0.983	0.982	0.984	0.983	0.991	0.992	0.990	0.991
ABDE	0.95	0.949	0.954	0.951	0.981	0.980	0.980	0.980	1	1	1	1
ABE	0.9083	0.911	0.911	0.911	0.966	0.970	0.966	0.968	0.991	0.992	0.991	0.991
AC	0.9875	0.986	0.988	0.987	0.987	0.986	0.988	0.987	1	1	1	1
ACD	0.8667	0.865	0.855	0.858	0.9	0.903	0.913	0.905	1	1	1	1
ACDE	0.875	0.883	0.881	0.880	0.95	0.952	0.952	0.952	0.962	0.966	0.966	0.966
ACE	0.9917	0.992	0.990	0.991	1	1	1	1	1	1	1	1
AD	1	1	1	1	1	1	1	1	1	1	1	1
ADE	1	1	1	1	1	1	1	1	1	1	1	1
AE	1	1	1	1	1	1	1	1	1	1	1	1
BC	0.9875	0.987	0.987	0.987	1	1	1	1	1	1	1	1
BCD	0.85	0.859	0.859	0.858	0.958	0.956	0.954	0.955	0.983	0.983	0.983	0.983
BCDE	0.8875	0.898	0.899	0.898	0.943	0.941	0.941	0.940	0.981	0.980	0.980	0.980
BCE	0.9833	0.984	0.984	0.984	0.991	0.987	0.993	0.990	0.983	0.984	0.982	0.982
BD	0.9875	0.987	0.988	0.987	1	1	1	1	1	1	1	1
BDE	1	1	1	1	1	1	1	1	1	1	1	1
BE	0.962	0.955	0.969	0.961	0.975	0.977	0.973	0.974	1	1	1	1
CD	0.737	0.737	0.737	0.731	0.925	0.924	0.924	0.924	0.975	0.977	0.973	0.974
CDE	0.891	0.892	0.893	0.892	0.966	0.965	0.962	0.963	0.966	0.969	0.964	0.965
CE	1	1	1	1	0.975	0.975	0.975	0.975	1	1	1	1
DE	1	1	1	1	1	1	1	1	1	1	1	1
Averaged	0.944	0.944	0.945	0.943	0.974	0.975	0.975	0.973	0.990	0.990	0.990	0.990

classifiers reached perfect accuracy, demonstrating the highest level of discriminative power for the main model. Even in more challenging multiclass combinations such as ABCDE and BCDE, SVM and DNN maintained accuracy above 96%, while RF and GB achieved performance levels in the range of 94–98%.

In terms of average accuracy, DNN and SVM reached 99%, representing the highest overall success. RF and GB achieved averages within the 97–98% range, whereas k-NN remained around 94%. These results

demonstrate that the hybrid feature set selected based on IG exhibited strong compatibility with different types of classifiers and outperformed the previous models in terms of both accuracy and stability. The consistent high performance of the main model in both binary and multiclass epileptic seizure detection scenarios highlight the potential adaptability of the proposed approach to clinical applications.

The ANOVA analysis presented in [Table 10](#) enabled a statistical comparison of the accuracy performances of Models 1–4 and the

Table 10

Statistical comparison of model accuracies using ANOVA across RF, SVM, k-NN, GB, and DNN classifiers.

No	F-statistic	P-value	Statistical Significance
RF	7.6321	$1.55 \cdot 10^{-5}$	Yes
SVM	13.8975	$2.09 \cdot 10^{-9}$	Yes
k-NN	15.4719	$2.63 \cdot 10^{-10}$	Yes
GB	5.3643	$5.09 \cdot 10^{-4}$	Yes
DNN	9.0985	$1.75 \cdot 10^{-6}$	Yes

IG-based main model under five different classifiers. The results indicated that there were statistically significant differences across all classifiers at $p < 0.001$ level. For the RF classifier, $F = 7.6321$ and $p = 1.55 \cdot 10^{-5}$ were obtained, demonstrating notable differences in the mean accuracies among the models. For SVM, $F = 13.8975$ and $p = 2.09 \cdot 10^{-9}$ yielded one of the strongest significance values, suggesting that SVM is highly sensitive to model variations. In k-NN, the highest F-statistics was observed with $F = 15.4719$ and $p = 2.63 \cdot 10^{-10}$, indicating that this method exhibited substantial accuracy variance across different model configurations. GB, with $F = 5.3643$ and $p = 5.09 \cdot 10^{-4}$, also demonstrated significant differences, although its effect was relatively lower compared to the other classifiers. For DNN, $F = 9.0985$ and $p = 1.75 \cdot 10^{-6}$ revealed a high level of significance, showing that the accuracy performance of the deep learning model can vary considerably depending on the employed feature set.

Overall, the consistently low p-values confirmed the presence of statistically significant accuracy differences between Models 1–4 and the main model across all classifiers. This finding supports that the hybrid feature set optimized through IG substantially influenced the classification performance of the model architectures and, in most cases, improved their accuracy.

These results provide the basis for a broader interpretation, which is elaborated in the following discussion section.

4. Discussion

In this study, the dataset used for epileptic seizure detection was specifically chosen due to its structural integrity, balanced sampling design, and widespread acceptance in the scientific literature. The EEG dataset from the Epileptology Department of the University of Bonn has a clear classification structure consisting of five distinct classes. These classes represent different states of consciousness (normal, interictal, ictal) and signals obtained from different brain regions, thereby covering a wide spectrum that includes both normal and pathological frequency components. In contrast, other publicly available datasets frequently used in the literature, such as CHB-MIT, generally consist of long-term clinical recordings, suffer from subject-dependency and incomplete annotations, or display imbalances in class distribution. The Bonn dataset, on the other hand, provides a more controlled and comparable framework through its high-resolution and segmented EEG recordings. Furthermore, its structure allows the formation of different class combinations, which enabled the IG-based feature selection analysis applied in this study to be evaluated under all class combinations. In this way, the classification performance could be tested not only on specific subclasses but also across all EEG signals, thereby improving the reliability of the results. For these reasons, the Bonn dataset has been considered the most appropriate choice for seizure detection studies, both in terms of methodological suitability and as a benchmark standard in literature.

The four different models used in this study reflect various dimensions of EEG signal characteristics. The first model represents statistical parameters in the time domain, while the second model captures spectral distributions in the frequency domain. The third model extracts hybrid features based on time–frequency analysis, and the fourth model incorporates linear prediction coefficients together with residual

components. The primary reason why a single model may be insufficient for epileptic seizure detection is that epileptic activity manifests itself across multiple domains, including time, frequency, and spectral energy distribution. Analyses relying on a single group of features may fail to capture the full set of characteristic signatures of seizures.

Therefore, selecting the most discriminative features of each model through the IG method and integrating them into a hybrid model significantly improved system performance. This approach preserves the strengths of each model while compensating for their weaknesses with the complementary features of the others. As a result, the final integrated model demonstrated more stable and higher accuracy rates in both binary and multi-class classification scenarios.

The choice of classifiers was also a strategic decision, aiming to leverage algorithms with different structural characteristics. Ensemble-based methods such as RF and GB achieved high performance with multidimensional and complex feature sets. SVM's provided high accuracy by sharply defining decision boundaries, particularly when using fewer features. The k-NN algorithm performed well in homogeneous subsets but showed relatively lower performance in highly complex multi-class combinations. The deep learning-based DNN captured complex and nonlinear relationships within the data, achieving the highest average accuracy. This result highlights the DNN's capability to effectively model the nonlinear and high-dimensional characteristics of epileptic seizures.

Although deep architectures such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based temporal models have shown promising results in EEG-based seizure detection, they were not adopted in the present study for specific methodological reasons. First, the Bonn dataset used in this research consists of short, single-channel EEG segments recorded under controlled conditions, which limits the spatial and temporal dependencies that CNNs and LSTMs are designed to exploit. Second, these architectures typically require large-scale, multi-channel, and continuous EEG datasets to prevent overfitting and to effectively learn hierarchical representations. The current study aimed to systematically evaluate the discriminative power of handcrafted statistical, spectral, and time–frequency features using interpretable and computationally efficient classifiers. This approach allows a clear quantitative comparison across 26 class combinations and provides direct insight into which features contribute most to seizure detection. Additionally, Temporal Transformers and hybrid deep architectures, while powerful, entail substantially higher computational complexity and parameterization, which are not optimal for the intended real-time or portable EEG applications. For these reasons, traditional machine learning models and a fully connected DNN were selected to balance interpretability, efficiency, and accuracy within the scope and data characteristics of this study.

Ensemble integration techniques such as stacking or blending were deliberately not applied in this study. Although these approaches can improve predictive accuracy, they may obscure the contribution of individual classifiers and reduce model interpretability. The comparative design adopted here was therefore preferred to ensure transparency and to evaluate each classifier's independent performance on the same hybrid feature set.

When compared to existing studies, the results demonstrate a remarkable superiority. The average accuracy of 99% obtained across 26 different class combinations surpasses many methods reported in the literature, providing high success not only in binary but also in multi-class classification tasks. Commonly used approaches such as DWT, FFT, WPD, and entropy-based methods are generally strong in specific feature domains but lack the generalizability of hybrid structures that integrate multiple feature spaces. The IG-based hybrid feature selection approach proposed in this study, therefore, provides superiority in terms of both accuracy and stability.

From a clinical perspective, the developed system can offer fast, accurate, and reliable results in epileptic seizure detection, effectively

supporting physicians in their decision-making process. The high accuracy rates reduce both false positives and false negatives, thereby preventing unnecessary medical interventions and minimizing diagnostic delays. This is particularly important for patient safety in ictal periods requiring urgent intervention. Moreover, the system's ability to perform both binary and multi-class classification enables not only the detection of seizure presence but also the differentiation of seizure types, contributing to the personalization of treatment plans.

Another important advantage of the proposed approach is its potential integration with wearable EEG devices. With continuous EEG monitoring made possible by such systems, the model could perform real-time analysis and send instant alerts to patients and healthcare teams. In this way, seizures occurring at home, during sleep, or during daily activities could be detected immediately, allowing rapid preventive actions. This feature could substantially improve clinical monitoring processes in epilepsy management.

As shown in Table 8 and Table 9, when the goal is solely to detect seizure occurrences (class combinations involving class E), the DNN classifier achieved an average accuracy of 99.69%. This result demonstrates the system's nearly flawless performance in ictal detection and indicates a critical level of reliability for clinical applications.

When the studies summarized in Table 11 are examined, it can be observed that high accuracy rates have been achieved in epileptic seizure detection using different feature extraction methods and classifiers. For example, Pattnaik et al. [33] and Harpale et al. [34] reported accuracy rates of 93–96% on the CHB-MIT dataset using wavelet transform-based features. Polat et al. [35] reported 82.5% accuracy on the Bonn dataset using time-domain features, while Eltrass et al. [36] achieved 97.8% accuracy on the same dataset with QKLMS-based energy features. Similarly, Chakraborty et al. [37] reported 98.2% accuracy using variational mode decomposition (VMD) and statistical features. More recent studies have adopted deep learning models; Liu et al. [38] achieved 98.94% accuracy with CNN-LSTM, Wang et al. [39] achieved 98.45% with CNN-VIT, and Abdulwahhab et al. [40] reported

99.57% accuracy using CNN/RNN architectures. A common characteristic of these studies is their primary focus on binary classification (seizure/non-seizure) scenarios and reporting results based on specific class combinations.

In contrast, the IG-based hybrid feature selection and multi-classifier approach proposed in this study goes beyond comparable works in the literature by evaluating all class combinations offered by the Bonn dataset. This comprehensive analysis was conducted on 26 different class combinations, achieving stable and high accuracy not only in binary but also in multi-class scenarios. In particular, the DNN classifier achieved an average accuracy of 99.69% in combinations involving class E, representing one of the highest performances reported for ictal detection. Furthermore, the IG-based hybrid approach not only enhanced accuracy but also improved model explainability by quantitatively identifying which features and class combinations contributed most to detection performance. Overall, this study addresses the gap in limited comparative analyses in literature and provides an important contribution to generalizability and clinical applicability in multi-class seizure detection scenarios.

Another important contribution to the proposed method is its potential for integration with mobile-based and wearable EEG systems. Many existing approaches for seizure detection have been evaluated only under laboratory conditions or through offline analyses, whereas the IG-based hybrid model developed here, due to its low computational cost in feature selection, is well-suited for real-time implementation on mobile devices. This capability enables continuous EEG monitoring and instantaneous seizure detection, thereby enhancing patient safety particularly in home environments or daily non-clinical settings. Furthermore, direct transmission of data obtained through mobile applications to physicians could support the personalization of treatment plans and improve follow-up processes. Hence, the integrability of the proposed approach with mobile health technologies introduces an important innovative dimension in epilepsy management.

Nevertheless, some limitations of the system should be acknowledged. The dataset used in this study consists of EEG recordings obtained in controlled laboratory settings from a limited number of participants. Artifacts commonly encountered in real clinical environments, such as muscle activity, eye-blink potentials, or electrode contact noise, are not represented in this dataset. In addition, the limited demographic diversity of the dataset may partially restrict generalizability across different age groups, genders, and clinical profiles. Therefore, future studies should focus on utilizing EEG recordings collected from larger populations, over longer durations, and in naturalistic environments to further improve the reliability and practical applicability of the system.

In addition, since the EEG data were recorded using a specific acquisition system in a controlled environment, minor variations could occur when signals are collected with different equipment. However, because the proposed hybrid model relies on relative spectral and statistical characteristics (such as normalized power spectral features, entropy, and Hjorth parameters), its structure is expected to remain robust across various EEG devices after appropriate calibration and normalization.

Considering these findings and interpretations, the concluding section summarizes the key contributions and outlines future directions.

5. Conclusions

This study presents a hybrid feature selection and classification approach that ensures high accuracy and reliability in epileptic seizure detection. Using the EEG dataset from the Epileptology Department of the University of Bonn, features extracted from time, frequency, and time–frequency domains were optimized with the IG method, and the strengths of four different models were integrated to form a hybrid structure. This hybrid feature set was systematically evaluated across 26 different class combinations, providing stable performance in both binary and multi-class classification scenarios. In this regard, the study

Table 11
Summary of reported EEG seizure detection methods across public datasets.

Ref.	Technique	Classifier	Dataset	Class	Accuracy (%)
[33]	Q-Wavelet Transform	SVM, RF	CHB-MIT	Seizure/ non-seizure	93.0
[34]	Pattern adaptive Wavelet Transform	Fuzzy	CHB-MIT	Seizure/ non-seizure	96.0
[35]	Time Features	SVM	Bonn	ABCDE	82.5
[36]	Signal Energy	QKLMS	Bonn	ABCDE	97.8
[37]	VMD, Kurtosis	RF	Bonn	AB-CD-E	98.2
[38]	DWT	SVM	Bonn	ADE	96.19
[39]	Periodic/aperiodic self-selection	CNN	CHB-MIT	Seizure/ non-seizure	98.94
[40]	CWT, Scaleogram	SVM	CHB-MIT	Seizure/ non-seizure	94.0
[41]	CWT, Phase & power Spectrograms	CNN-VIT	CHB-MIT	Seizure/ non-seizure	98.45
[42]	Time & Frequency Features	DTW Matching	CHB-MIT	Seizure/ non-seizure	98.90
[43]	Periodic/aperiodic PSD parameterization	Classic Machine Learning	Bonn	10 Different Classification Task	98.88
[44]	STFT, CWT	CNN, RNN	Bonn	Seizure/ non-seizure	99.57
[45]	Mobile Net IG	k-NN	CHB-MIT	Seizure/ non-seizure	97.40
[46]	Fuzzy IG	DRNN	Not Specified	Preictal/ interictal	97.87
[47]	Feature importance	XGboost	Bonn	Seizure/ non-seizure	Not Specified

addresses the gap of limited comparative analyses in the literature and enhances generalizability in multi-class contexts.

The results obtained demonstrated nearly flawless performance in ictal detection, particularly with the DNN classifier, which achieved an average accuracy of 99.69% in class combinations including class E. Clinically, this is of great importance for ensuring high sensitivity and low error rates in epilepsy diagnosis. Furthermore, the IG-based hybrid feature selection quantitatively identified which features and class combinations contributed most to detection performance, thereby enhancing the explainability of the model and strengthening its role as a design guideline.

The clinical applicability of the developed system should also be emphasized. High accuracy and low false positive/negative rates effectively support physicians in the decision-making process, reducing unnecessary interventions and diagnostic delays. The potential integration with wearable EEG devices enables real-time seizure detection and alert systems in home environments or daily life, significantly improving patient safety.

Nevertheless, certain limitations must be acknowledged. The dataset used in this study was collected under controlled laboratory conditions and from a limited number of participants, which may restrict the system's performance in real clinical settings. Future research should incorporate long-term EEG recordings from diverse demographic groups, noisy field data, and transfer learning-based approaches to further enhance the system's generalizability and practical applicability. Investigating the real-time feasibility of the hybrid model and optimizing different class combinations also remain important research directions.

In conclusion, the proposed IG-based hybrid feature selection and multi-classifier approach makes a significant contribution to the literature on epileptic seizure detection. It stands out as a promising method in terms of accuracy, explainability, generalizability, and clinical applicability.

CRedit authorship contribution statement

Nuri Ikizler: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gunes Ekim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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